

UAV and AI Application for Runway Foreign Object Debris (FOD) Detection

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Abstract— There are several ways in which Foreign Object Debris (FOD) are detected on runways. Some of these methods include Radar, infrared technologies, and stationary cameras mounted on the runway and use image processing tools to find these FODs. Radar technology is highly accurate when finding FODs but is highly inaccurate with small FOD items causing a high false-positive rate. Stationary based RGB camera-based methods also have a high false-positive rate prompting the shutdown of runways, creating disruptions for both the airport and the airline carriers. The paper presents a new method of detection by using an Unmanned Aerial Vehicle (UAV) to fly above the runway at a low altitude (e.g. < 30 m) to find FOD in. We developed a system that combines a UAV, an RGB camera, an Artificial Intelligence (AI) detector trained using deep learning methods and locally collected images over a runway. The classes specifically looked at were paper, metal, bolts, plastic, and plastic bottles. Different lighting conditions of both full sunlight and cloudy weather were taken into consideration when the images were collected. The detector was trained with various data augmentation techniques including resize, rotate, and colour augmentation. Results have concluded that there is a potential use for UAV's as a method of FOD detection, with a high rate of accuracy in the detections. This could lead to shorter timeframes and fewer disruptions where runways are closed.

Keywords—Foreign Object Debris, FOD, Runway, Debris Detection, Unmanned Aerial Vehicle.

I. INTRODUCTION

Within the aviation industry, foreign object debris is any item or substance that is considered alien to an aircraft system. These alien substances can potentially cause damage and disrupt operations of aircraft. There are two types of FOD, external and internal. External FOD relates to hazards such as bird strikes, objects on the runway, hail, ash-clouds, nuts, bolts, rubber, stone, plastic and even personal belongings. Internal FOD relates to hazards such as items left within the aircraft which interfere with flight safety (cables tangled, inoperable parts etc).

FOD at airports can cause damage that costs airports and airlines millions of dollars every year with the estimated cost to the aerospace industry being upwards of \$4 billion a year [1]. The responsibility of the clearing of these runways falls to both the airport and the airlines, with the recommendation by ICAO stating that "*The surface of pavements (runways, taxiways, aprons, etc.) should be kept clear of any loose stones or other objects that might cause damage to airplane structures or engines, or impair the operation of airplane systems.*"[2]

On runways or taxiways, maintenance for FOD is carried out using several methods:

- Magnetic Bars which are suspended under truck or cars which inspect the runway.
- Sweeping, which is carried out manually with an airfield sweeper.
- Rumble strips used to dislodge FOD from vehicle undercarriages, so they are not brought onto highly active taxi ways and runways.
- FOD containers placed at all gates for the collection of debris for the collection of rubbish and plastics.
- Cameras used to view runways from maintenance locations or the tower.
- Millimeter wave radars which continuously scan runway surfaces.

Running these methods are costly, as the prices for equipment, staff and training of staff are significantly high [3]. The cost also of the closure of a runway also needs to be taken into consideration, with some airports in Europe stating that 1% of their delays being due to FOD [4]. Airports can also close the runways because of false positives from current methods of FOD detection, such as radar [5] and thermal wave imaging.

Several FOD detection techniques have been proposed. These include the use of video-based detection from cameras mounted on the side of the runway [6], wavelet analysis of images [7] [8], convolutional neural networks [9] [10] [11], wireless sensor networks [12], and millimeter-wave radar [13]. The design and applications of UAVs for surveillance and environmental monitoring has exponentially increased over the past 15 years [14-17]. This paper presents a new method of detection using a UAV and AI, to detect FOD at low altitude (< 30 m).

II. CONCEPT OF OPERATIONS

The concept of the paper is that a UAV will be able to fly from a base on one end of the runway to a second base on the opposite end. During the flight the UAV will take video footage of the runway and process the data searching for FOD's that may be on the runway. This can be seen in fig 1 where the UAV is able to search the width of the runway ensuring the quickest flight time, were with a small UAV

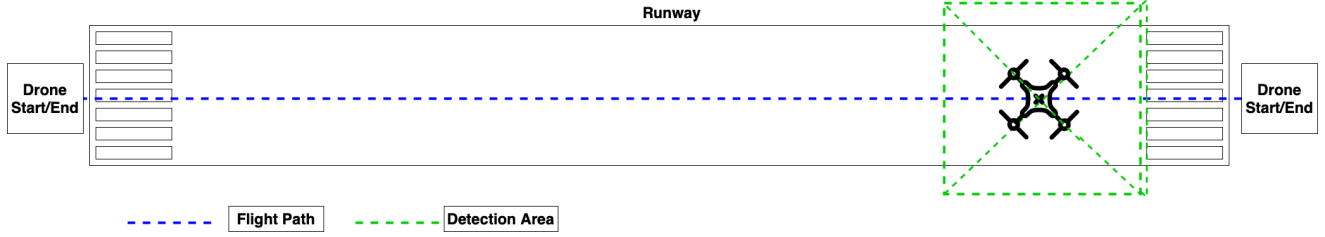


Fig. 1. Graphical interpretation of CONOPS

such as the DJI Phantom and the DJI spark used in this project takes 183-270 seconds to cover a 49 m x 3300 m long runway. Table I summarizes the time it will take for different types of off the shelf UAV's to cover the runway in one flight depending on the type of camera they have installed and its field of view. During this process the UAV will identify all possible FOD and relay the size of the object back to the control tower or once returning to the docking station on the opposite end of the runway, the UAV will recharge until next deployment.

TABLE I. SUMMARY OF FLIGHT TIME VS UAV CAMERA FIELD OF VIEW

	DJI					
Model	Phantom 3	Mavic Air	Spark	Phantom 4 Pro	Mavic Pro	Mavic 2 Pro
Sensor	1/2.3" CMOS	1/2.3" CMOS	1/2.3" CMOS	1" CMOS	1/2.3" CMOS	1" CMOS
Sensor Size	6.16x4.62p	6.17x4.55p	6.17x4.55p	13.2x8.8p	6.17x4.55p	13.2x8.8p
Aspect Ratio	4:3	4:3	4:3	3:2	4:3	3:2
Lens	f/2.8	f/2.8	f/2.6	f/2.8	f/2.2	f/2.8
Local Length	20mm	24mm	25mm	24mm	28mm	28mm
Image Size	4000x3000	4000x3000	3960x2976	4000x3000	4000x3000	4000x3000
FOV	94°	85°	81.9°	84°	78.8°	77°
Height Needed To Fly	39.6m	46.3m	48.9m	51.1m	51.7m	57.8m
Flight Time Needed	4.16 min	7.8 min	25 min	3.2 min	3.8 min	6.25 min

A DJI Spark was used for data collection [18] and testing however similar UAVs can be used. A basic system architecture is shown in fig 2a and 2b.

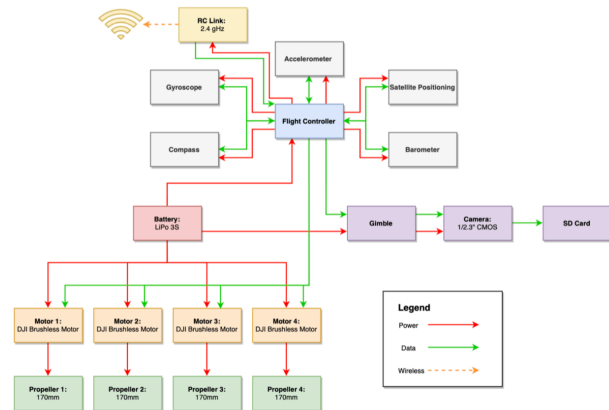


Fig. 2a. System Architecture of the DJI Spark – Airborne Elements

The system architecture of the DJI spark consists of an

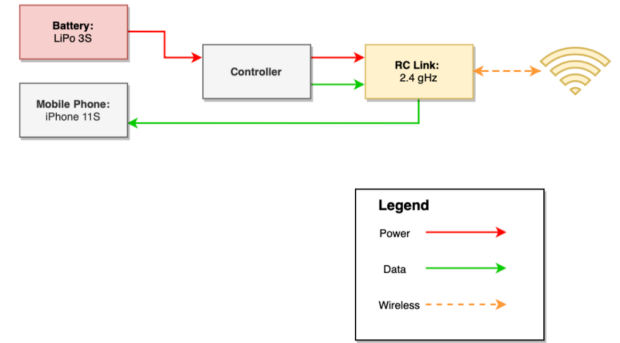


Fig. 2b. System Architecture of the DJI Spark – Ground Elements

airborne element (fig 2a) containing an inbuilt camera and flight controller, and a ground element (fig 2b) which uses the user's mobile device to connect to the UAV via a WIFI link set up by a separate DJI controller.

III. METHODS AND TOOLS

A. Models

Two methods were explored for FOD detection, Yolov3 [19] and Microsoft Azure Custom Vision [20]. The same data set was used on both platforms and the two were trained on the same data sets. The better performing model as then used to complete a full-scale test.

Yolov3 uses a single neural network applied to an entire image. The neural network divides the images up into regions and then predicts the bounding boxes and probabilities. The probabilities are predicted by pre-determined weights which is one of the many features to Yolov3 [19]. Due to its ability to look at the whole image the predictions are created from the global context of an image. Yolov3 currently boasts increased speeds over classic models such as R-CNN (1000 times faster) and Fast R-CNN (100 times faster). Yolov3 currently has 80 different classes trained within its model which can be applied to any photo. The model also gives the ability to train personal data sets by retraining the top three layers of the classifier and detector. Yolov3 has pre trained models for certain subject material to allow the user to build an optimised model. These fields are limited with the examples of landmarks, fruit, retail items and animals.

The Microsoft Azure Custom Vision toolbox allows building, deploying and improving image classifiers. The Custom Vision functionality can be used in two different ways; image classification, applying one or more labels to an image, and object detection which returns the coordinates in

the image of the classes [20]. The Custom Vision service by Microsoft is optimised to recognise differences in the images which it is provided. It was possible to train the model on as little as 50 images per label before getting a significant mAP result. When the model was trained on a general subject material, the Microsoft AI system finds the most appropriate training model for that specific data set giving a higher chance of resulting in a better model.

B. Data Set Creation

The training and validation data sets were created from four different locations. The four locations consisted of two car parks which were cemented with a dark grey and another two car parks cemented with a light grey. These two contrasting colours aim to replicate the colour of the runways and taxiways found at most airports. The type of weather was also taken into consideration with most images taken on both sunny and partially cloudy days.

Each of the classes in the data set has images taken at various heights. These heights range between 3 and 25 meters to ensure a large data set which has a large variation was created. Five classes, paper, plastic, bottle, metal and metal bolts were considered to demonstrate the methods.

The first data set that was created was Paper. This was chosen as it is common to have small pieces of paper which may fall onto the runway. Although not particularly destructive if ingested, this class was added as an example of FOD that could be found on the runway. A piece of plain white paper was used to represent this class in the images.

The second data set was Plastic. This data set was created due to the large volumes of plastic that can be blown onto the runway, from tape to rubbish. These can not only come from both the hangers, warehouses and airport facilities beside the runway but also the terminal itself. Two small black pipes were used to represent the plastic class.

The Bottle data set was created as a separate data set from the plastic as bottles tend to have a common shape and a larger mass than small pieces of plastic.

Medium sized pieces of metal were used to create a metal class representing the pieces of metal that could fall off airliners while taking off or landing. This type of debris is one of the most common to cause major issues if ingested by an engine. Small pieces of metal were bought from a local hardware store and used to represent this class.

Metal bolts were separated into a class of their own. This was due to the size of the bolts being smaller than the medium sized pieces of metal. Two types of bolts were purchased from a local hardware store, one gold plated steel and the other general steel sized 30 to 60 mm long. These were used to represent the bolts class.

1500 images were collected for each class all varying in height and location. Each of these images are labelled for the training process of the model. At each of the locations a secondary validation set of images were taken at specific heights of 3, 6, 12, 20 and 25 meters. This was used to determine if there was a bias to a specific height within the trained model as will be discussed in section V.

C. Image Labelling

The labelling technique is different for the two models. Labellmg is used to label the images for the YoloV3 training. Labellmg is a graphical image annotation tool, which allows labelling objects using bounding boxes in images. Labellmg is a simple process in which each image that has been captured can be opened and labelled with a specific class. As seen in fig 3, a bounding box can be placed over each specific class in an image even where multiple classes may be found. These locations on the image can be saved in the appropriate

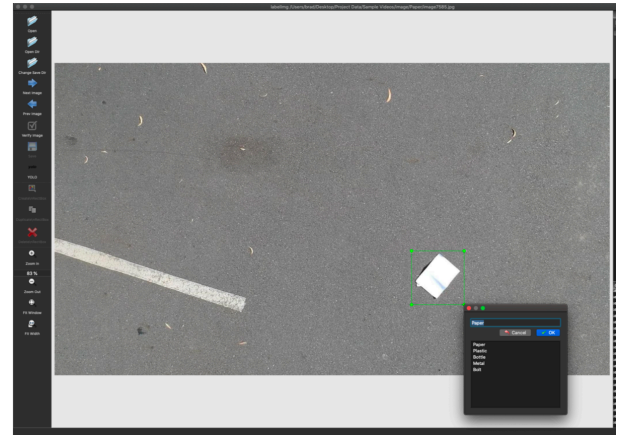


Fig. 3. Labelling process within Labellmg

format to be used in training the classifier.

The Microsoft Custom Vision labelling system has a simpler user interface. Once the images are uploaded there is a tagged and untagged option which allows the user to know which images have been labelled and which have not. The process requires the user to click on the image that needs to be labelled and draw a bounding box over the top. This allows the user to either create or choose an existing label to label

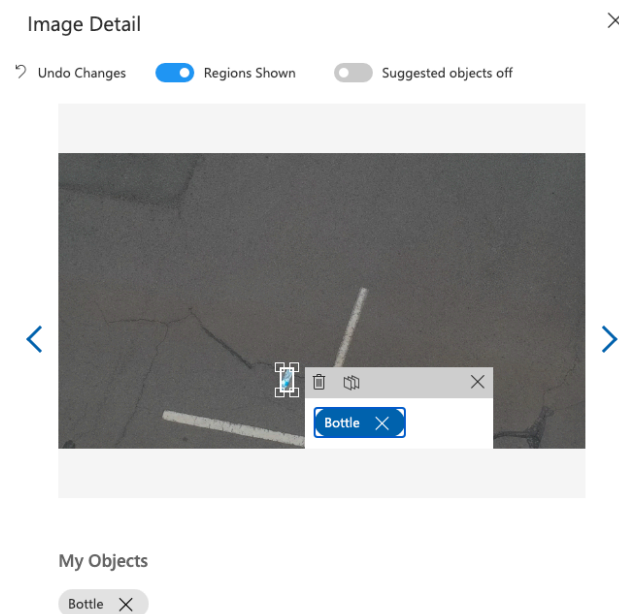


Fig. 4. Labelling process in Microsoft Azure Custom Vision

that bounding box as seen in fig 4.

The two main benefits of using the Microsoft Azure Custom Vision over YoloV3 is that when images are loaded into the Custom Vision, if there is an object that can clearly be identified, the labelling system will automatically place a dashed bounding box around the object giving the user a quicker way to label each of the items. The second benefit is that after labelling around 100 images of each class the model can be "Quick trained". This quick training method provides the option of suggested objects to the user.

The "Suggested objects" variable passes through each of the untagged images, finds the object in the image and places one of the classes over it. Using these suggestions, the user has the option to accept the bounding box location and the associated class, or just the class and readjust the bounding box to fit the object correctly.

D. Detection Model 1 – YOLOv3

Yolov3 was used by incorporating Google Collaborate [19]. This enables the use of a high-powered GPU on a virtual machine which helps decrease the training time, while being able to be run constantly without the main computer running a CPU or GPU at 100%.

Google Collaborate provides the optimal environment to apply the classification learner for Yolov3. A GPU was used within the notebook allowing GPU acceleration using the Yolov3 system. This allows the processing to process detections over one hundred times faster than using a general CPU. The hardware accelerator of the notebook was then required to be changed to the GPU to ensure it was being used correctly.

Google Collaborate offers the ability to use Google drive in sync with any code used in the Collaborate notebook. This enables the ability to place all images and the labelled files into a folder on a google drive and link it to the Collaborate notebook. This was done by mounting the associated google drive into the cloud VM.

AlexeyAB's repository [19], was cloned to build the darknet repository. This repository was specifically used with Yolov3 and was required to complete the training process of the model [19]. An adjustment was created to the Makefile to enable OPENCV and GPU use for darknet. The Darknet repository was then built.

Custom configuration files are created from this repository to fit the number of images being used and the number of classes. The .cfg file was altered to allow for the needs of the object detector. The code was opened in text editor and the following changes were made: The batch was changed to 64 and subdivisions = 16 for ultimate results. This can be scaled down if issues arise. Other sections of code were changed based on the number of classes that are being used. Max batches was set to 2000 times the number batches, and the steps are 80% and 90% of the max batches. The classes are changed to 5 as there were five classes being trained for in the three Yolo layers. The filters were changed to 30 in the three convolutional layers before the YOLO layers. If there was a memory issue all the random values within the document would be set to 0. This would however reduce the accuracy of the model, so random was kept at 1.

Two new files were created stating the class names and the split of the data set into testing and training.

The next step was to download the weights for the convolutional layers of the YOLOv3 network. This helped the model have a higher accuracy and ensured a faster training time. Once completed a list of file names was used to create a training file which holds the locations of all the images in the Darknet folder.

Once the setup was complete the detector can be trained using one single line of code. The training process takes up to 28 hours due to the large number of images in the data set. Using Yolov3 eliminated the need to manually create images that had been augmented as within the Yolov3 programming the options to change augmentation is available. This data augmentation happens during the training of the model itself, as it completed the augmentation based on settings the user applies within the cfg file. The saturation, exposure, size, grain and hue of an image was augmented within the Yolov3 framework.

One of the main limitations to the use of google collaborate is the run-time that is allowed by each user. Due to the GPU use being shared among users, Google Collaborate only allows a run-time of up to 12 hours at a time. This means that in each instance, the training model will be interrupted at least once. This requires a backup line of code to be written in the event that training is cut short. This single line of code takes the last trained weights and continues from on from that point. The model was then able to be used within Google Collaborate for validation with the validation images.

E. Detection Model 2 – Microsoft Azure Custom Vision

A button called train is available 24/7 during the labelling process which offers both quick training and advanced training. Quick training as mentioned above in section III.D was used during the labelling process. The full training method was able to give a high-level indication on how each of the classes have trained and how the overall model precision, recall and mAP is. Using this data, a clear indication of where the model lacks can be seen and changed progressively to obtain a more precise model.

Advanced training is a complete training process of the model. This uses Microsoft's AI to find the appropriate training for this specific set of data while also providing the most appropriate data augmentation within the model training itself. The advanced training allows the users to determine how long to train their model from one to twenty-four hours. On the free azure account training is free, however the model takes longer to train due to the processing power being limited. In any already existing Azure account, the training is also free if the account is specifically set up with Microsoft. This model was given a training budget of 24 hours to allow the longest possible training. The training was given processing priority through the Microsoft cloud services. This meant the model had completed training within 3 hours as it was given a high priority for training.

The mAP, Recall and Precision values are automatically displayed within Custom Vision. This allows the ability to not only see the entire model's performance but the

performance of each tag and how the tag affected the model as a whole. This can be used to alter the data sets to create a more accurate model.

IV. EXPERIMENTS

The two models, YOLOv3 and Microsoft Azure Custom Vision were used to evaluate in an experimental capacity to find the appropriate height at which each of the model are best suited for FOD detection. As each model was trained by two different methods, YOLOv3 and Microsoft Custom Vision, the height which is optimal for each observation changes. Once the base height was chosen each model was then used to find FOD in a full-scale test.

A. Test Sites

Two test sites were considered. The first site, which was used to test the height accuracy of the model, had no links to the locations where the training data sets were created. This allowed no influence in the results of the model.



Fig. 5. Carpark site used for base experiments

The second site, used for the full-scale testing, was an inactive paved glider runway, located in South East Queensland, Australia. This simulated the general types of runways. It also provided neutral ground to test the model without any bias towards a specific area or class.

B. Experimental Setup

Multiple objects from each of the classes were laid out in small area. At each of the locations, leaves, rocks and other small debris were left within the field of view to clutter the area, testing the model's accuracy. The UAV was flown over the cluttered area at specific heights while recording a video. The heights at which the UAV was flown were at; 3 meters, 6 meters, 12 meters, 20 meters and 25 meters. These video recordings were then processed through both the YOLOv3 model and the Microsoft Custom Vision model and results were recorded. These models were then compared to find which has a higher accuracy within all classes. This higher accuracy model was used for the full-scale testing at the second location.

In the full-scale testing multiple objects from each class were placed in the area within close proximity of each other to test the model's ability to distinguish items that have some similar features, such as the Metal and Bolts classes. A video



Fig. 6. Glider runway used for full-scale testing

was taken on the UAV at the specific height required for the final model. This was then processed by the model and results are scored to determine if FOD detection is possible. Items were then separated to a larger distance and video was processed with the items apart.

V. RESULTS

A. Base Experimental Results

The results evaluate the model when data sets are created using a large array of heights. The performance of the object identification was higher, even if misidentified, in the Custom Vision model when compared to the YOLOv3 model. Using this data, for either of the models to obtain enough detail from each of the classes to identify them, the UAV with the current camera will have to fly below 12 meters. This would require the UAV to do four flights across the runway due to the camera limitations of the DJI spark.

The accuracy of the YOLOv3 model over each of the heights varies largely where most of the true positives are found where the UAV was flown at $< 3\text{m}$. When the UAV was flown at higher altitudes there are more false positives and even no indication of any objects in the field of view. Table II shows the results of each height, how many classes were in each of the images, the number of correctly identified classes, the number of incorrectly identified classes and the accuracy of the model at each height.

TABLE II. OVERVIEW OF THE YOLOV3 MODEL RESULTS

UAV Height (m)	Classes in Image	Objects In Image	Classes Detected	Objects Identified Correctly	Missed Identifications	Accuracy of Model
3	3	3	3	3	0	100%
3	2	3	2	3	0	100%
6	3	5	3	1	0	20%
6	2	3	1	0	1	0%
12	4	11	0	0	0	N/A
20	5	11	0	0	0	N/A
25	5	11	1	0	1	0%

For the Microsoft Azure Custom Vision model there was a higher accuracy among all of the heights. The highest accuracy was at 12 meters, where out of the 11 objects that

were placed in the area, 12 objects were correctly identified, and 1 object was miss identified. Table III shows the accuracy results of the model at all heights.

TABLE III. OVERVIEW OF THE MICROSOFT AZURE CUSTOM VISION MODEL RESULTS

UAV Height (m)	Classes in Image	Objects In Image	Classes Detected	Objects Identified Correctly	Missed Identifications	Accuracy of Model
3	3	3	3	3	0	100%
3	2	3	2	2	1	66%
6	3	5	3	4	1	80%
6	2	3	1	0	1	0%
12	4	11	3	10	1	91%
20	5	11	2	1	5	9%
25	5	11	1	0	5	0%

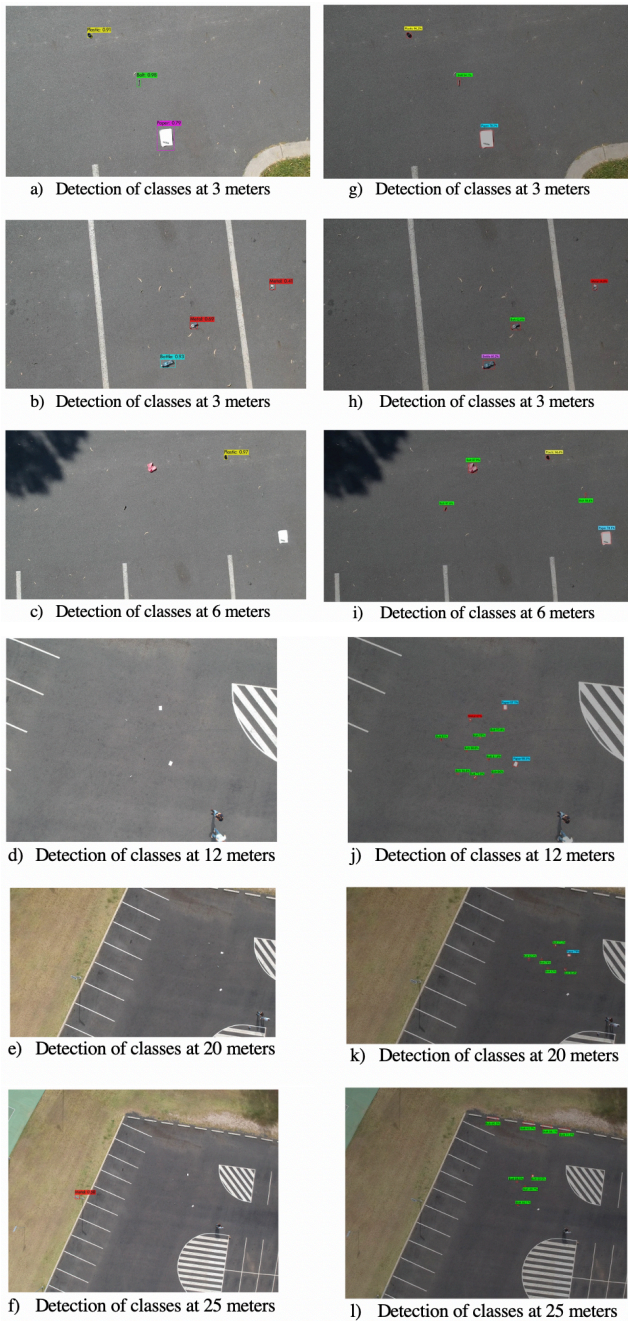


Fig. 7. Detection of classes at different heights using Yolo a) to f) and Microsoft Azure Custom Vision model g) to l).

The two models, YOLOv3 and Microsoft Azure Custom Vision were compared with one another to determine which model would be the most appropriate for the FOD detection. Comparisons were also made to the base model of YOLOv3 which helps indicate whether the model created through YOLOv3 is also at an acceptable level.

Fig 8 shows that the Mean Average Precision of the Microsoft Azure Custom Vision was significantly higher than those models created using YOLOv3. The mAP is at 82.9% whereas the YOLO models range from 28.2% to 33%. Due to the YOLOv3 models having such a low mAP the Microsoft Azure Custom Vision model was used for the full-scale testing.

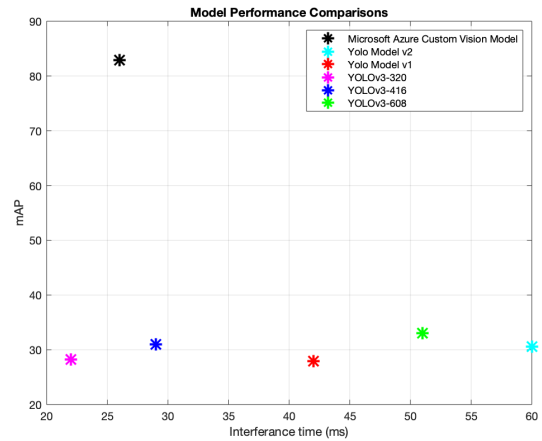


Fig. 8. ROC curve of the model with each class

B. Full-Scale Testing

In the full-scale testing, the final images were evaluated through the Microsoft Azure Custom Vision model to find objects that were placed out on the runway. The model was able to detect not only all of the objects within the image, but also able to correctly identify each of the classes. Even with the limitation of the camera specifications on the DJI Spark, each of the detections have an accuracy of above 70%. The Paper class had a probability of 91.7%, Bottle at 78%, Plastic at 76.4%, Metal strips of which there were two at 72.4% and 83.1%, and finally the Metal Bolts class which were detected with a probability of 85% and 97.4%. These results can be



Fig. 9. Full-scale testing model results with objects in close proximity

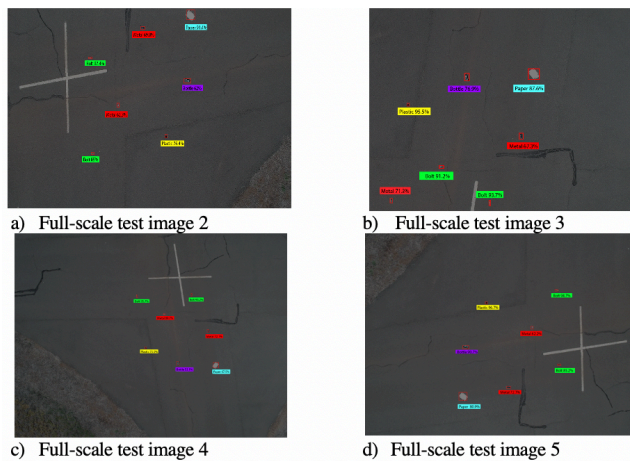


Figure 10. Full-scale test images with randomly placed objects results

seen in fig 9 and 10 where the first full-scale test had each of the object within close proximity of each other.

The second test consisted of object which were separated. This resulted in the model identifying each of the objects with a high probability. Fig 11 shows each of the classes being found in this second test. The Paper class was detected at 98.1%, Bottle at 89.1%, Metal at 80.5%, Bolts at 93.9% for the gold-plated bolt and 81.8% for the steel bolt, and Plastic at 76.2% probability. The images also show that there are no false positives detected even when the model is set at a low probability threshold. Each of the images for testing were taken while the UAV was moving at 1 meter per second. This allowed a seamless simulation of what would be received during proper deployment.

Fig 12 shows that the ROC curve of the model is significantly high with an area of 0.89. The model is able to decipher both the Paper (0.97) and the Metal class (0.97), with the area under the curve (AUC) being large. The class

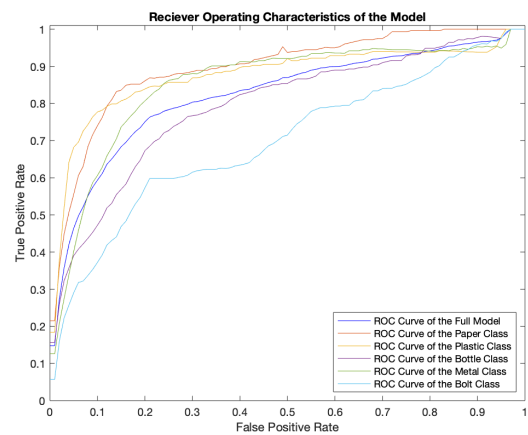


Fig. 12. ROC curve of each class and the complete model

the model had most difficulties with was the Bolt class, with its AUC being 0.72. This is due to the detailed elements of the bolts not being recognised as correctly as the other classes. This is a known limitation as bolts are small objects which can vary easily be missed depending on the height of the UAV being flown and the camera being used. Also due to the small in nature size of the bolts (30mm to 60mm), there is also a higher percentage that the model will label other objects as bolts giving them a high percentage of accuracy for that class. These false positives become greater as the UAV if flown higher. Another class that the model had difficulties with is the bottle class. This class has an AUC of 0.73 which is slightly higher than the bolts class. Another possible reason for the Bottle class to have a low AUC is the amount of reflected sunlight in each of the data set images. This reflected light could distort the features of the bottle. Quite often the bottles in the testing phase were labelled by the

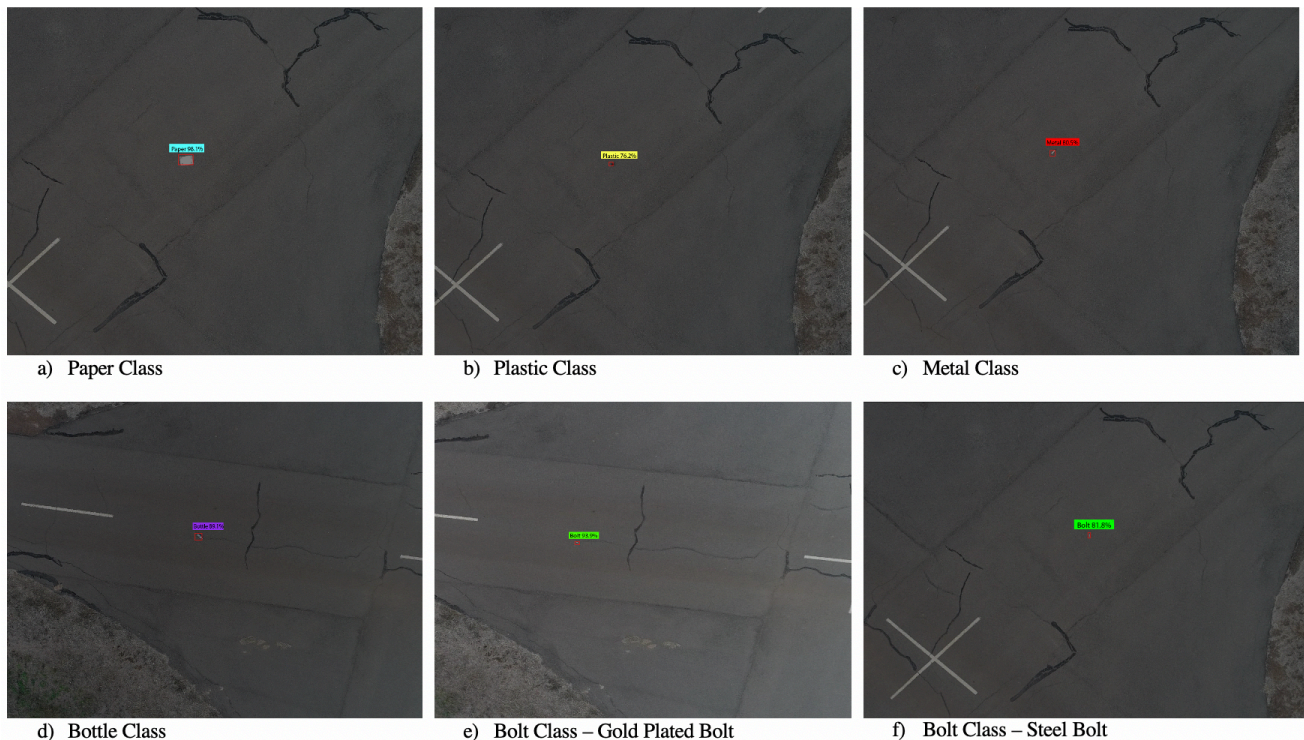


Fig. 11. Full-scale testing results with object spread over the runway

model as bolts. However, at the height of 12m where the model excels, this false positive did not arise.

VI. CONCLUSIONS

This paper proposed a method which combines using a UAV and AI for FOD detection on runways.

Data sets of 1500 images for five different classes, Paper, Plastic, Bottle, Metal and Bolts, were acquired from 4 separate sites. Two data model were explored, YoloV3 and Microsoft Custom Vision. Each data set was labelled in the specifications of two models to enable training for each model. The training of each model included data augmentation on each of the data sets which were automatically created by the training program.

Full-scale testing showed that Microsoft Custom Vision model was able to detect each of the FOD classes laid out in its field of view and correctly class each of the objects. The ability to retrain the model to improve the mAP with more data gives greater model adaptability for multiple situational types, included the addition of further FOD classes if required. The final test shows promise into the possibility of future use on runways. Testing of not only the model but the particular UAV that was used gives greater understanding as to how the UAV will need to be set up and configured for the best possible practice of FOD detection.

A video illustrating the system can be found under this link: <https://youtu.be/mc7UdTRf5aI>

ACKNOWLEDGEMENTS

The authors would like to acknowledge Peter Trotter and Aspect UAV Imaging for the use of a DJI Mavic Pro and finding a suitable runway for full-scale testing, and George Maren for the use of the inactive glider runway.

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